

A fully Bayesian model decision for per-individual mixture-effect SNPs:

BF of “one nonzero π ” (pure) vs. “ ≥ 2 nonzero π ” (mixture)

derivation note for CHIMERA / SuSiE-Mix

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Setup. Causal SNP, genotype $g_i \in \{0, 1, 2\}$, codings $x_i^A = g_i$, $x_i^D = \mathbf{1}\{g_i \geq 1\}$, $x_i^R = \mathbf{1}\{g_i = 2\}$, $m \in \{A, D, R\}$. Latent responder type $z_i \in \{A, D, R\}$, $z_i \mid \boldsymbol{\pi} \sim \text{Cat}(\boldsymbol{\pi})$, $\boldsymbol{\pi} = (\pi_A, \pi_D, \pi_R) \in \Delta^2$. Effects $\beta_A, \beta_D, \beta_R$.

$$y_i \mid z_i=m, \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(x_i^m \beta_m, \sigma^2), \quad p(y_i \mid \boldsymbol{\pi}, \boldsymbol{\beta}, \sigma^2) = \sum_m \pi_m \mathcal{N}(x_i^m \beta_m, \sigma^2). \quad (1)$$

1 The decision, stated as an event on the simplex

The scientific question is exactly a partition of the prior support of $\boldsymbol{\pi}$:

$$H_{\text{pure}} : \text{ exactly one } \pi_m > 0 \quad (\boldsymbol{\pi} \in \{\mathbf{e}_A, \mathbf{e}_D, \mathbf{e}_R\}), \quad H_{\text{mix}} : \text{ at least two } \pi_m > 0.$$

We want the Bayes factor $\text{BF} = \frac{p(\mathbf{y} \mid H_{\text{mix}})}{p(\mathbf{y} \mid H_{\text{pure}})}$. Place priors

$$\beta_m \mid \sigma^2 \sim \mathcal{N}(0, \sigma^2 g_0) \quad (m \in \{A, D, R\}), \quad \sigma^2 \sim \text{Inv-Gamma}(\frac{\nu_0}{2}, \frac{\nu_0 s_0^2}{2}), \quad \boldsymbol{\pi} \sim \text{Dirichlet}(\boldsymbol{\alpha}). \quad (2)$$

The key technical move: *marginalise the latent $\mathbf{z}$ exactly*. Because the two hypotheses are events for $\mathbf{z}$ ($H_{\text{pure}} : \mathbf{z}$ constant; $H_{\text{mix}} : \geq 2$ distinct values), and because conditional on $\mathbf{z}$ both $\boldsymbol{\beta}$ and $\boldsymbol{\pi}$ integrate out in closed form, the whole computation reduces to a sum over partitions with closed-form summands.

2 Exact marginal likelihood as a sum over partitions

A configuration $\mathbf{z} \in \{A, D, R\}^n$ defines groups $G_m(\mathbf{z}) = \{i : z_i = m\}$ with counts $\mathbf{n}(\mathbf{z}) = (n_A, n_D, n_R)$.

(i) **Integrate $\boldsymbol{\pi}$ (Dirichlet–multinomial).**

$$P(\mathbf{z}) = \int_{\Delta^2} \text{Dirichlet}(\boldsymbol{\pi}; \boldsymbol{\alpha}) \prod_i \pi_{z_i} d\boldsymbol{\pi} = \frac{B(\boldsymbol{\alpha} + \mathbf{n}(\mathbf{z}))}{B(\boldsymbol{\alpha})} = \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + n)} \prod_m \frac{\Gamma(\alpha_m + n_m)}{\Gamma(\alpha_m)}, \quad \alpha_0 = \sum_m \alpha_m. \quad (3)$$

(ii) **Integrate β (and σ^2) given $\mathbf{z}$.** Given $\mathbf{z}$, the likelihood factorises across groups, each a Bayesian linear regression of $\mathbf{y}_{G_m}$ on the coding- m column. With σ^2 known and $\beta_m \sim \mathcal{N}(0, \sigma^2 g_0)$, writing $u_m = x_{G_m}^m$, $a_m = u_m^\top u_m$, $b_m = u_m^\top \mathbf{y}_{G_m}$,

$$\log \text{ML}_m(G_m) = -\frac{1}{2} \left[n_m \log(2\pi\sigma^2) + \log(1 + g_0 a_m) + \frac{1}{\sigma^2} (\mathbf{y}_{G_m}^\top \mathbf{y}_{G_m} - \frac{g_0 b_m^2}{1 + g_0 a_m}) \right]. \quad (4)$$

(Verified against direct numerical integration over β to 10^{-13} .) For unknown σ^2 under the prior (2),

replace (4) by the multivariate- t marginal: pooling all groups, $p(\mathbf{y} | \mathbf{z}) = \pi^{-n/2} \frac{\Gamma(\frac{\nu_0+n}{2})}{\Gamma(\frac{\nu_0}{2})} \frac{(\nu_0 s_0^2)^{\nu_0/2}}{(\det M_{\mathbf{z}})^{1/2} (\nu_0 s_0^2 + \text{RSS}_{\mathbf{z}})}$ with $M_{\mathbf{z}}$ the block-diagonal $(I + g_0 U^\top U)$ over groups and $\text{RSS}_{\mathbf{z}} = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top U (g_0^{-1} I + U^\top U)^{-1} U^\top \mathbf{y}$.

(iii) **Assemble.** With $p(\mathbf{y} | \mathbf{z}) = \prod_m \text{ML}_m(G_m)$,

$$p(\mathbf{y} | H) = \frac{\sum_{\mathbf{z} \in H} P(\mathbf{z}) p(\mathbf{y} | \mathbf{z})}{\sum_{\mathbf{z} \in H} P(\mathbf{z})}, \quad \text{BF} = \frac{\sum_{\mathbf{z}: \geq 2 \text{ groups}} P(\mathbf{z}) p(\mathbf{y} | \mathbf{z}) / \sum_{\mathbf{z}: \geq 2} P(\mathbf{z})}{\sum_m P(\mathbf{z} = m\mathbf{1}) p(\mathbf{y} | \mathbf{z} = m\mathbf{1}) / \sum_m P(\mathbf{z} = m\mathbf{1})} \quad (5)$$

The three “pure” configurations $\mathbf{z} = m\mathbf{1}$ are the partitions with one occupied group; for those $p(\mathbf{y} | m\mathbf{1}) = \text{ML}_m(\{1, \dots, n\})$ is exactly the standard single-coding SER marginal likelihood. Thus H_{pure} is the three single-coding SERs and H_{mix} is every non-degenerate partition; (5) is the entire model decision, with no maximisation anywhere.

3 Computation

The exact sum has 3^n terms (tractable only as a unit test; the formulas above were checked this way). For data-scale inference use any of:

- **Collapsed Gibbs over $\mathbf{z}$:** sample $z_i | \mathbf{z}_{-i}$ with β, π integrated out (a Chinese-restaurant-style update using (3)–(4)); estimate $p(\mathbf{y} | H_{\text{mix}})$ by Chib’s method or bridge sampling.
- **Mean-field variational Bayes** on $(\mathbf{z}, \beta, \pi, \sigma^2)$ with factorised $q(\mathbf{z}) \prod_m q(\beta_m) q(\pi) q(\sigma^2)$. The ELBO is a rigorous lower bound on $\log p(\mathbf{y} | H_{\text{mix}})$; the updates are the Bayesian analogue of the EM in the package and run at SER cost. *This is the SuSiE-compatible route:* the per-individual mixture becomes the inner SER, and IBSS propagates it to full fine-mapping.
- **Laplace** about the VB mode for a fast BF estimate (note label-switching: restrict to a canonical ordering or use the collapsed form).

4 Why the Bayesian decision is well-posed (and self-regularising)

1. **No boundary pathology.** The frequentist LRT for $\pi_D = \pi_R = 0$ is non-regular (unidentified β_D, β_R under the null; Hartigan). Here we *integrate* over π and β , so BF in (5) is a ratio of finite, well-defined marginal likelihoods. The non-regularity simply does not arise.
2. **Automatic Occam penalty.** H_{mix} spreads its prior mass over many partitions; when the data are explained by a single coding, the non-degenerate partitions contribute little and $\text{BF} \lesssim 1$. No explicit complexity penalty is needed.

3. **Favouring additive is a prior choice, made explicit.** Two knobs control it: (a) the effect-prior scale g_0 (the BF is genuinely sensitive to it—use a proper, scale-matched g_0 , never improper); (b) an asymmetric $\alpha = (\alpha_A, \alpha_D, \alpha_R)$ with α_A large to concentrate the pure prior at the additive vertex, and/or a spike-and-slab mixing the vertex prior (weight $1 - \rho$) with Dirichlet on the interior (weight ρ). Then posterior odds $= \rho/(1 - \rho) \cdot \text{BF}$.

5 Asymptotics: this BF is detectable at realistic n

As $n \rightarrow \infty$ with truth P^* generated by (1), $\frac{1}{n} \log \text{BF} \rightarrow \text{KL}(P^* \parallel \mathcal{M}_{\text{pure}}) = \min_m \text{KL}(P^* \parallel \text{pure coding } m)$, the KL to the *best single coding*. Crucially each pure model has a single free coefficient, so it cannot absorb either the dominance curvature or the within-genotype overdispersion of a true mixture; hence $\text{KL}(P^* \parallel \mathcal{M}_{\text{pure}}) = \Theta(\theta^2)$ (with $\theta = \beta/\sigma$), *not* the $\Theta(\theta^4)$ of a saturated-genotypic comparison. Required sample size (numerically verified KL; GWAS threshold $n^* \approx 19.8/\text{KL}$):

$\theta = \beta/\sigma$	0.10	0.20	0.30
KL(mix $\parallel$ best pure)	3.8×10^{-4}	4.1×10^{-3}	2.2×10^{-2}
n^* (genome-wide power)	$\approx 5.2 \times 10^4$	$\approx 4.8 \times 10^3$	$\approx 9 \times 10^2$

So the *pure-vs-mixture* BF is well within reach: $n^* \sim 10^3$ for strong eQTLs and $\sim 10^4$ – 10^5 for weak ones—feasible in GTEx and trivially so in biobanks. (Contrast: a mixture-vs-saturated genotypic BF, which is what one needs to exclude a single deterministic dominance variant, is $\Theta(\theta^4)$ and requires $n \sim 10^6$.)

6 Recommended model set

The BF (5) answers “is the SNP a single pure coding, or not?”. A positive result is necessary but not sufficient to claim a *per-individual* mixture, because a single deterministic genotypic (2-df dominance) effect, or two distinct causal variants each pure, can also defeat H_{pure} . The clean remedy is a finite Bayesian model set with the same closed-form ingredients:

$$\{\text{pure-}A, \text{pure-}D, \text{pure-}R, \text{genotypic (2-df)}, \text{mixture}\},$$

each scored by its marginal likelihood, and posterior model probabilities formed by normalising. The genotypic member is (4) with the 2-column design $[x^D \mid x^R]$; the multiple-distinct-variant alternative is handled natively by SuSiE’s multiple single effects. The per-individual-mixture claim is the event that the *mixture* model wins against *all* of these—which is the honest, decision-complete version of the test.